

2.1 Statements and Quantifiers



Figure 2.2 Construction of a logical argument, like that of a house, requires you to begin with the right parts. (credit: modification of work “Barn Raising” by Robert Stinnett/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Identify logical statements.
2. Represent statements in symbolic form.
3. Negate statements in words.
4. Negate statements symbolically.
5. Translate negations between words and symbols.
6. Express statements with quantifiers of all, some, and none.
7. Negate statements containing quantifiers of all, some, and none.

Have you ever built a club house, tree house, or fort with your friends? If so, you and your friends likely started by gathering some tools and supplies to work with, such as hammers, saws, screwdrivers, wood, nails, and screws. Hopefully, at least one member of your group had some knowledge of how to use the tools correctly and helped to direct the construction project. After all, if your house isn't built on a strong foundation, it will be weak and could possibly fall apart during the next big storm. This same foundation is important in logic.

In this section, we will begin with the parts that make up all logical arguments. The building block of any logical argument is a **logical statement**, or simply a statement. A logical statement has the form of a complete sentence, and it must make a claim that can be identified as being true or false.

When making arguments, sometimes people make false claims. When evaluating the strength or validity of a logical argument, you must also consider the **truth values**, or the identification of true or false, of all the statements used to support the argument. While a false statement is still considered a logical statement, a strong logical argument starts with true statements.

Identifying Logical Statements



Figure 2.3 Not all roses are red. (credit: “assorted pink yellow white red roses macro” by ProFlowers/Flickr, CC BY 2.0)

An example of logical statement with a false truth value is, “All roses are red.” It is a logical statement because it has the form of a complete sentence and makes a claim that can be determined to be either true or false. It is a false statement because not all roses are red: some roses are red, but there are also roses that are pink, yellow, and white. Requests, questions, or directives may be complete sentences, but they are not logical statements because they cannot be determined to be true or false. For example, suppose someone said to you, “Please, sit down over there.” This request does not make a claim and it cannot be identified as true or false; therefore, it is not a logical statement.

EXAMPLE 2.1

Identifying Logical Statements

Determine whether each of the following sentences represents a logical statement. If it is a logical statement, determine whether it is true or false.

1. Tiger Woods won the Master’s golf championship at least five times.
2. Please, sit down over there.
3. All cats dislike dogs.

Solution

1. This is a logical statement because it is a complete sentence that makes a claim that can be identified as being true or false. As of 2021, this statement is true: Tiger Woods won the Master’s in 1997, 2001, 2002, 2005 and 2019.
2. This is not a logical statement because, although it is a complete sentence, this request does not make a claim that can be identified as being either true or false.
3. This is a logical statement because it is a complete sentence that makes a claim that can be identified as being true or false. This statement is false because some cats do like some dogs.

YOUR TURN 2.1

Determine whether each of the following sentences represents a logical statement. If it is a logical statement, determine whether it is true or false.

1. The Buffalo Bills defeated the New York Giants in Super Bowl XXV.
2. Michael Jackson’s album *Thriller* was released in 1982.
3. Would you like some coffee or tea?

Representing Statements in Symbolic Form

When analyzing logical arguments that are made of multiple logical statements, **symbolic form** is used to reduce the amount of writing involved. Symbolic form also helps visualize the relationship between the statements in a more

concise way in order to determine the strength or validity of an argument. Each logical statement is represented symbolically as a single lowercase letter, usually starting with the letter p .

To begin, you will practice how to write a single logical statement in symbolic form. This skill will become more useful as you work with compound statements in later sections.

EXAMPLE 2.2

Representing Statements Using Symbolic Form

Write each of the following logical statements in symbolic form.

1. Barry Bonds holds the Major League Baseball record for total career home runs.
2. Some mammals live in the ocean.
3. Ruth Bader Ginsburg served on the U.S. Supreme Court from 1993 to 2020.

Solution

1. p : Barry Bonds holds the Major League Baseball record for total career home runs. The statement is labeled with a p . Once the statement is labeled, use p as a replacement for the full written statement to write and analyze the argument symbolically.
2. q : Some mammals live in the ocean. The letter q is used to distinguish this statement from the statement in question 1, but any lower-case letter may be used.
3. r : Ruth Bader Ginsburg served on the U.S. Supreme Court from 1993 to 2020. When multiple statements are present in later sections, you will want to be sure to use a different letter for each separate logical statement.

YOUR TURN 2.2

Write each of the following logical statements in symbolic form.

1. The movie *Gandhi* won the Academy Award for Best Picture in 1982.
2. Soccer is the most popular sport in the world.
3. All oranges are citrus fruits.

WHO KNEW?

Mathematics is not the only language to use symbols to represent thoughts or ideas. The Chinese and Japanese languages use symbols known as Hanzi and Kanji, respectively, to represent words and phrases. At one point, the American musician Prince famously changed his name to a symbol representing love.

[BBC Prince Symbol Article \(https://openstax.org/r/magazine-36107590\)](https://openstax.org/r/magazine-36107590)

Negating Statements

Consider the false statement introduced earlier, “All roses are red.” If someone said to you, “All roses are red,” you might respond with, “Some roses are not red.” You could then strengthen your argument by providing additional statements, such as, “There are also white roses, yellow roses, and pink roses, to name a few.”

The **negation of a logical statement** has the opposite truth value of the original statement. If the original statement is false, its negation is true, and if the original statement is true, its negation is false. Most logical statements can be negated by simply adding or removing the word *not*. For example, consider the statement, “Emma Stone has green eyes.” The negation of this statement would be, “Emma Stone does not have green eyes.” The table below gives some other examples.

Logical Statement	Negation
Gordon Ramsey is a chef.	Gordon Ramsey is not a chef.
Tony the Tiger does not have spots.	Tony the Tiger has spots.

Table 2.1

The way you phrase your argument can impact its success. If someone presents you with a false statement, the ability to rebut that statement with its negation will provide you with the tools necessary to emphasize the correctness of your position.

EXAMPLE 2.3**Negating Logical Statements**

Write the negation of each logical statement in words.

1. Michael Phelps was an Olympic swimmer.
2. Tom is a cat.
3. Jerry is not a mouse.

✓ Solution

1. Add the word *not* to negate the affirmative statement: "Michael Phelps was not an Olympic swimmer."
2. Add the word *not* to negate the affirmative statement: "Tom is not a cat."
3. Remove the word *not* to negate the negative statement: "Jerry is a mouse."

> YOUR TURN 2.3

Write the negation of each logical statement in words.

1. Ted Cruz was not born in Texas.
2. Adele has a beautiful singing voice.
3. Leaves convert carbon dioxide to oxygen during the process of photosynthesis.

Negating Logical Statements Symbolically

The symbol for negation, or not, in logic is the tilde, $\sim$. So, not p is represented as $\sim p$. To negate a statement symbolically, remove or add a tilde. The negation of not (not p) is p . Symbolically, this equation is $\sim(\sim p) = p$.

EXAMPLE 2.4**Negating Logical Statements Symbolically**

Write the negation of each logical statement symbolically.

1. p : Michael Phelps was an Olympic swimmer.
2. r : Tom is not a cat.
3. $\sim q$: Jerry is not a mouse.

✓ Solution

1. To negate an affirmative logical statement symbolically, add a tilde: $\sim p$.
2. Because the symbol for this statement is r , its negation is $\sim r$.
3. The symbol for this statement is $\sim q$, so to negate it we simply remove the $\sim$, because $\sim(\sim q) = q$. The answer is q .

> YOUR TURN 2.4

Write the negation of each logical statement symbolically.

1. $\sim p$: Ted Cruz was not born in Texas.
2. q : Adele has a beautiful voice.
3. r : Leaves convert carbon dioxide to oxygen during the process of photosynthesis.

Translating Negations Between Words and Symbols

In order to analyze logical arguments, it is important to be able to translate between the symbolic and written forms of logical statements.

EXAMPLE 2.5

Translating Negations Between Words and Symbols

Given the statements:

r : Elmo is a red Muppet.

p : Ketchup is not a vegetable.

1. Write the symbolic form of the statement, "Elmo is not a red Muppet."
2. Translate the statement $\sim p$ into words.

Solution

1. "Elmo is not a red muppet" is the negation of "Elmo is a red muppet," which is represented as r . Thus, we would write "Elmo is not a red muppet" symbolically as $\sim r$.
2. Because p is the symbol representing the statement, "Ketchup is not a vegetable," $\sim p$ is equivalent to the statement, "Ketchup is a vegetable."

YOUR TURN 2.5

Given the statements:

r : Woody and Buzz Lightyear are best friends.

$\sim p$: Wonder Woman is not stronger than Captain Marvel.

1. Write the symbolic form of the statement, "Wonder Woman is stronger than Captain Marvel."
2. Translate the statement $\sim r$ into words.

Expressing Statements with Quantifiers of All, Some, or None

A **quantifier** is a term that expresses a numerical relationship between two sets or categories. For example, all squares are also rectangles, but only some rectangles are squares, and no squares are circles. In this example, *all*, *some*, and *none* are quantifiers. In a logical argument, the logical statements made to support the argument are called **premises**, and the judgment made based on the premises is called the **conclusion**. Logical arguments that begin with specific premises and attempt to draw more general conclusions are called **inductive arguments**.

Consider, for example, a parent walking with their three-year-old child. The child sees a cardinal fly by and points it out. As they continue on their walk, the child notices a robin land on top of a tree and a duck flying across to land on a pond. The child recognizes that cardinals, robins, and ducks are all birds, then excitedly declares, "All birds fly!" The child has just made an inductive argument. They noticed that three different specific types of birds all fly, then synthesized this information to draw the more general conclusion that all birds can fly. In this case, the child's conclusion is false.

The specific premises of the child's argument can be paraphrased by the following statements:

- Premise: Cardinals are birds that fly.
- Premise: Robins are birds that fly.
- Premise: Ducks are birds that fly.

The general conclusion is: "All birds fly!"

All inductive arguments should include at least three specific premises to establish a pattern that supports the general conclusion. To counter the conclusion of an inductive argument, it is necessary to provide a counter example. The parent can tell the child about penguins or emus to explain why that conclusion is false.

On the other hand, it is usually impossible to prove that an inductive argument is true. So, inductive arguments are considered either strong or weak. Deciding whether an inductive argument is strong or weak is highly subjective and often determined by the background knowledge of the person making the judgment. Most hypotheses put forth by scientists using what is called the “scientific method” to conduct experiments are based on inductive reasoning.

In the following example, we will use quantifiers to express the conclusion of a few inductive arguments.

EXAMPLE 2.6

Drawing General Conclusions to Inductive Arguments Using Quantifiers

For each series of premises, draw a logical conclusion to the argument that includes one of the following quantifiers: all, some, or none.

1. Squares and rectangles have four sides. A square is a parallelogram, and a rectangle is a parallelogram. What conclusion can be drawn from these premises?
2. $1 + 2 = 3$, $6 + 7 = 13$, and $23 + 24 = 47$. Of these, 1 and 2, 6 and 7, and 23 and 24 are consecutive integers; 3, 13, and 47 are odd numbers. What conclusion can be drawn from these premises?
3. Sea urchins live in the ocean, and they do not breathe air. Sharks live in the ocean, and they do not breathe air. Eels live in the ocean, and they do not breathe air. What conclusion can be drawn from these premises?

Solution

1. The conclusion you would likely come to here is “Some four-sided figures are parallelograms.” However, it would be incorrect to say that all four-sided figures are parallelograms because there are some four-sided figures, such as trapezoids, that are not parallelograms. This is a false conclusion.
2. From these premises, you may draw the conclusion “All sums of two consecutive counting numbers result in an odd number.” Most inductive arguments cannot be proven true, but several mathematical properties can be. If we let n represent our first counting number, then $n + 1$ would be the next counting number and $n + (n + 1) = 2n + 1$. Because $2n$ is divisible by two, it is an even number, and if you add one to any even number the result is always an odd number. Thus, the conclusion is true!
3. Based on the premises provided, with no additional knowledge about whales or dolphins, you might conclude “No creatures that live in the ocean breathe air.” Even though this conclusion is false, it still follows from the known premises and thus is a logical conclusion based on the evidence presented. Alternatively, you could conclude “Some creatures that live in the ocean do not breathe air.” The quantifier you choose to write your conclusion with may vary from another person’s based on how persuasive the argument is. There may be multiple acceptable answers.

YOUR TURN 2.6

For each series of premises, draw a logical conclusion to the argument that includes one of the following quantifiers: all, some, or none.

1. $1 + 2 = 3$, $5 + 6 = 11$, and $14 + 15 = 29$. Of these, 1 and 2 are consecutive integers, 5 and 6 are consecutive integers, and 14 and 15 are consecutive integers. Also, their sums, 3, 11, and 29 are all prime numbers. Prime numbers are positive integers greater than one that are only divisible by one and the number itself. What conclusion can you draw from these premises?
2. A robin is a bird that lays blue eggs. A chicken is a bird that typically lays white and brown eggs. An ostrich is a bird that lays exceptionally large eggs. If a bird lays eggs, then they do not give live birth to their young. What conclusion can you draw from these premises?
3. All parallelograms have four sides. All rectangles are parallelograms. All squares are rectangles. What additional conclusion can you make about squares from these premises?

 It is not possible to prove definitively that an inductive argument is true or false in most cases.

Negating Statements Containing Quantifiers

Recall that the negation of a statement will have the opposite truth value of the original statement. There are four basic forms that logical statements with quantifiers take on.

- All A are B .

- Some A are B .
- No A are B .
- Some A are not B .

The negation of logical statements that use the quantifiers *all*, *some*, or *none* is a little more complicated than just adding or removing the word *not*.

For example, consider the logical statement, "All oranges are citrus fruits." This statement expresses as a subset relationship. The set of oranges is a subset of the set of citrus fruit. This means that there are no oranges that are outside the set of citrus fruit. The negation of this statement would have to break the subset relationship. To do this, you could say, "At least one orange is not a citrus fruit." Or, more concisely, "Some oranges are not citrus fruit." It is tempting to say "No oranges are citrus fruit," but that would be incorrect. Such a statement would go beyond breaking the subset relationship, to stating that the two sets have nothing in common. The negation of " A is a subset of B " would be to state that " A is not a subset of B ," as depicted by the Venn diagram in [Figure 2.4](#).

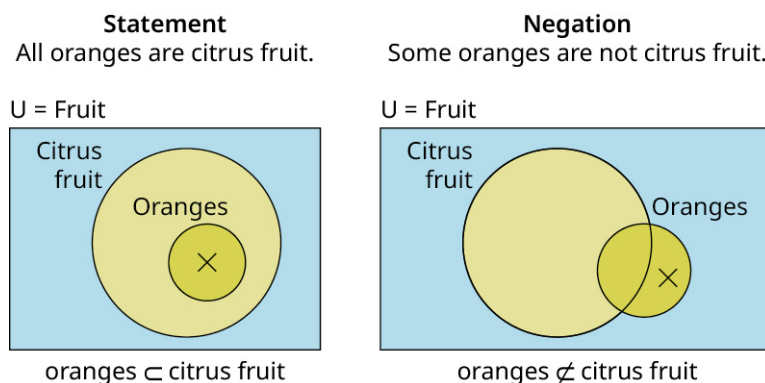


Figure 2.4

The statement, "All oranges are citrus fruit," is true, so its negation, "Some oranges are not citrus fruit," is false.

Now, consider the statement, "No apples are oranges." This statement indicates that the set of apples is disjoint from the set of oranges. The negation must state that the two are not disjoint sets, or that the two sets have a least one member in common. Their intersection is not empty. The negation of the statement, " A intersection B is the empty set," is the statement that " A intersection B is not empty," as depicted in the Venn diagram in [Figure 2.5](#).

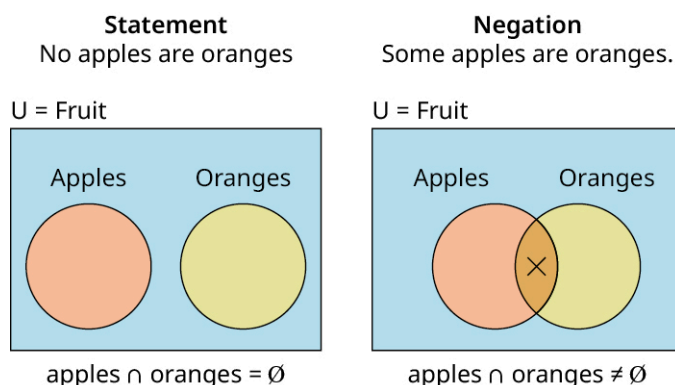


Figure 2.5

The negation of the true statement "No apples are oranges," is the false statement, "Some apples are oranges."

[Table 2.2](#) summarizes the four different forms of logical statements involving quantifiers and the forms of their associated negations, as well as the meanings of the relationships between the two categories or sets A and B .

Logical Statements with Quantifiers	Negation of Logical Statements w/Quantifiers
Form: All A are B . Means: A is a subset of B , $A \subset B$. All zebras have stripes. (True)	Form: Some A are not B . Means: A is not a subset of B , $A \not\subset B$. Some zebras do not have stripes. (False)
Form: Some A are B . Means: A intersection B is not empty, $A \cap B \neq \emptyset$. Some fish are sharks. (True)	Form: No A are B . Means: A intersection B is empty, $A \cap B = \emptyset$. No fish are sharks. (False)
Form: No A are B . Means: A intersection B is empty, $A \cap B = \emptyset$. No trees are evergreens. (False)	Form: Some A are B . Means: A intersection B is not empty, $A \cap B \neq \emptyset$. Some trees are evergreens. (True)
Form: Some A are not B . Means: A is not a subset of B , $A \not\subset B$. Some horses are not mustangs. (True)	Form: All A are B . Means: A is a subset of B , $A \subset B$. All horses are mustangs. (False)

Table 2.2

We covered sets in great detail in [Chapter 1](#). To review, " A is a subset of B " means that every member of set A is also a member of set B . The intersection of two sets A and B is the set of all elements that they share in common. If A intersection B is the empty set, then sets A and B do not have any elements in common. The two sets do not overlap. They are disjoint.

 VIDEO

[Logic Part 1A: Logic Statements and Quantifiers \(https://openstax.org/r/Logic_Statements_and_Quantifiers\)](https://openstax.org/r/Logic_Statements_and_Quantifiers)

EXAMPLE 2.7

Negating Statements Containing Quantifiers *All*, *Some*, or *None*

Given the statements:

p : All leopards have spots.

r : Some apples are red.

s : No lemons are sweet.

Write each of the following symbolic statements in words.

1. $\sim p$
2. $\sim r$
3. $\sim s$

 Solution

1. The statement "All leopards have spots" is p and has the form "All A are B ." According to [Table 2.2](#), the negation will have the form "Some A are not B ." The negation of p is the statement, "Some leopards do not have spots."
2. The statement "Some apples are red" has the form "Some A are B ." This indicates that the categories A and B overlap or intersect. According to [Table 2.2](#), the negation will have the form, "No A are B ," indicating that A and B do not intersect. This results in the opposite truth value of the original statement, so the negation of "Some apples are red" is the statement: "No apples are red."
3. Because s is the statement: "No lemons are sweet," s is asserting that the set of lemons does not intersect with the set of sweet things. The negation of s , $\sim s$, must make the opposite claim. It must indicate that the set of lemons intersects with the set of sweet things. This means at least one lemon must be sweet. The statement, "Some lemons are sweet" is $\sim s$. The negation of the statement, "No A are B ," is the statement, "Some A are B ," as indicated in [Table 2.2](#).

> YOUR TURN 2.7

Given the statements:

$\sim p$: Some apples are not sweet.

r : No triangles are squares.

s : Some vegetables are green. Write each of the following symbolic statements in words.

1. p
2. $\sim r$
3. $\sim s$

Check Your Understanding

1. A _____ is a complete sentence that makes a claim that may be either true or false.
2. The _____ of a logical statement has the opposite truth value of the original statement.
3. If p represents the logical statement, "Marigolds are yellow flowers," then _____ represents the statement, "Marigolds are not yellow flowers."
4. The statement $\sim(\sim p)$ has the same truth value as the statement _____.
5. The logical statements used to support the conclusion of an argument are called _____.
6. _____ arguments attempt to draw a general conclusion from specific premises.
7. All, some, and none are examples of _____, words that assign a numerical relationship between two or more groups.
8. The negation of the statement, "All giraffes are tall," is _____.



SECTION 2.1 EXERCISES

For the following exercises, determine whether the sentence represents a logical statement. If it is a logical statement, determine whether it is true or false.

1. A loan used to finance a house is called a mortgage.
2. All odd numbers are divisible by 2.
3. Please, bring me that notebook.
4. Robot, what's your function?
5. In English, a conjunction is a word that connects two phrases or parts of a sentence together.
6. $8 - 3 = 5$.
7. $7 + 3 = 11$.
8. What is 7 plus 3?

For the following exercises, write each statement in symbolic form.

9. Grammy award winning singer, Lady Gaga, was not born in Houston, Texas.
10. Bruno Mars performed during the Super Bowl halftime show twice.
11. Coco Chanel said, "The most courageous act is still to think for yourself. Aloud."
12. Bruce Wayne is not Superman.

For the following exercises, write the negation of each statement in words.

13. Bozo is not a clown.
14. Ash is Pikachu's trainer and friend.
15. Vanilla is the most popular flavor of ice cream.
16. Smaug is a fire breathing dragon.
17. Elephant and Piggy are not best friends.
18. Some dogs like cats.
19. Some donuts are not round.
20. All cars have wheels.
21. No circles are squares.
22. Nature's first green is not gold.
23. The ancient Greek philosopher Plato said, "The greatest wealth is to live content with little."

24. All trees produce nuts.

For the following exercises, write the negation of each statement symbolically and in words.

25. p : Their hair is red.
 26. $\sim q$: My favorite superhero does not wear a cape.
 27. s : All wolves howl at the moon.
 28. t : Nobody messes with Texas.
 29. $\sim u$: I do not love New York.
 30. $\sim v$: Some cats are not tigers.
 31. $\sim q$: No squares are not parallelograms.
 32. $\sim p$: The President does not like broccoli.

For the following exercises, write each of the following symbolic statements in words.

33. Given: p : Kermit is a green frog; translate $\sim p$ into words.
 34. Given: $\sim r$: Mick Jagger is not the lead singer for The Rolling Stones; translate r into words.
 35. Given: q : All dogs go to heaven; translate $\sim q$ into words.
 36. Given: $\sim s$: Some pizza does not come with pepperoni on it; translate s into words.
 37. Given: $\sim p$: No pizza comes with pineapple on it; translate $\sim(\sim p)$ into words.
 38. Given: r : Not all roses are red; translate $\sim(\sim r)$ into words.
 39. Given: $\sim t$: Thelonious Monk is not a famous jazz pianist; translate $\sim(\sim t)$ into words.
 40. Given: $\sim v$: Not all violets are blue; translate $\sim(\sim v)$ into words.

For the following exercises, draw a logical conclusion from the premises that includes one of the following quantifiers: all, some, or none.

41. The Ford Motor Company builds cars in Michigan. General Motors builds cars in Michigan. Chrysler builds cars in Michigan. What conclusion can be drawn from these premises?
 42. Michelangelo Buonarroti was a great Renaissance artist from Italy. Raphael Sanzio was a great Renaissance artist from Italy. Sandro Botticelli was a great Renaissance artist from Italy. What conclusion can you draw from these premises?
 43. Four is an even number and it is divisible by 2. Six is an even number and it is divisible by 2. Eight is an even number and it is divisible by 2. What conclusion can you draw from these premises?
 44. Three is an odd number and it is not divisible by 2. Seven is an odd number and it is not divisible by 2. Twenty-one is an odd number and it is not divisible by 2. What conclusion can you draw from these premises?
 45. The odd number 5 is not divisible by 3. The odd number 7 is not divisible by 3. The odd number 29 is not divisible by 3. What conclusion can you draw from these premises?
 46. Penguins are flightless birds. Emus are flightless birds. Ostriches are flightless birds. What conclusion can you draw from these premises?
 47. Plants need water to survive. Animals need water to survive. Bacteria need water to survive. What conclusion can you draw from these premises?
 48. A chocolate chip cookie is not sour. An oatmeal cookie is not sour. An Oreo cookie is not sour. What conclusion can you draw from these premises?